Facilitator guide: You Are the Riverton Health Department

For Jackie Massuda · Recitation for "Where Did You Get Sick?" (lecture Oct 23, 2026) · Eric Delmelle

Goal. Groups act as a health department and receive evidence about one fictional case in four rounds, committing to a decision each time. The aim is not to find the cause. It is to notice how interpretation and confidence change as evidence accumulates, and to practice deciding before the uncertainty goes away. Students saw the same town in lecture; here they work with case-level detail, must commit as a group, and face public pressure.

Before the session

Form groups of 4–5. Print one set per group: page 1 (briefing) with the record sheet (page 7), then Rounds 1, 2, 3 and 4 (pages 2–6) as separate handouts. Hand out each round only after the group has written its decision for the previous one. If printing is a problem, project the round pages instead.

Setup, roles	Round 1	Round 2	Round 3	Round 4 + final	Debrief
5 min	8 min	8 min	9 min	13 min	10 min

Say out loud: decide before the next round, with no going back; there is no hidden answer, so do not confirm or reject any group's explanation. If a group asks a factual question, answer only from *Release if asked* below; otherwise reply "the department doesn't know that yet; would you collect it?"

Round 1 · Where the cases live

Purpose: a visible pattern invites a confident explanation. *Expect:* "something in the water", "a factory", "coincidence", "older people"; most groups choose B or C. *Prompts:* What would make this pattern *not* unusual? What would choosing A right now cost?

Release if asked: at national rates (about 25 per 100,000 per year) a town of 18,000 would expect roughly 22 diagnoses in five years, so the town total is ordinary and only the concentration is unusual. The Flats is somewhat older than the town average (don't explain why until Round 4). Smoking, jobs, and residential history are not in the registry.

Round 2 · A possible source

Purpose: a plausible source makes the story feel solved. *Expect:* a jump toward A (test the soil, warn residents). *Prompts:* Distance is not exposure; how would a non-worker actually take these chemicals in? The site has been capped since 2009 and homes use reservoir water, so which routes are still open today? What does soil testing cost in fear, even if the results come back clean?

Release if asked: residential yard soil has never been sampled, and results would take about two months. There are no air measurements from the years the plant operated.

Round 3 · Where people worked and spent time

Purpose: the link may run through jobs rather than homes; a second plant appears; a shared place can mislead. *Expect:* a shift toward an occupational explanation, and interest in the senior center. *Prompts:* If the cause was at work, does warning Flats residents make sense? Which former workers are not on this map at all? The senior center: these are older neighbors; can a place they visit now explain a cancer that develops over decades? (Contrast with Sam in the lecture, where a shared place *would* matter for an infection.)

Release if asked: 13 of the 23 have a smoking history, roughly typical for people with this cancer. Keller employed several hundred people at its peak; a retirees' association still has about 150 members. Ashby's employment records are held by the company that bought it.

Round 4 · Time and residential history

Purpose: timing reshapes the cluster. Some of today's Flats cases arrived recently (senior housing); some of the people most plausibly exposed have moved away and now sit "outside" the cluster.

Expect: confidence drops; some groups move from A to B or C, others keep A for former workers. The *Meanwhile* page pushes toward visible action. *Prompts:* Whom does the 2021–2025 map miss? Is the problem now "the Flats", "Keller", "two workplaces", or several things at once? Which action is justified even if residential exposure turns out not to matter (for example, contacting former workers)? What do you say at Thursday's meeting if you are not sure?

Debrief (whole class)

Tally each group's final decision (A–D) on the board, then choose three or four:

- At what round did you have enough to act? What was the risk of acting one round earlier, or one round later?
- What happens if you act and the threat was not real? If you wait and it was?
- How much information about people's movements should a health department be allowed to collect?
- Could the same action protect one group and harm another (a public warning about the Flats, for the residents of Riverside Commons or for homeowners)?
- Who should decide when the evidence is strong enough: scientists, health officials, elected leaders, residents?

Close with the lecture's question: *when do we know enough to act?*

Avoid presenting the case key as the solution, turning the rounds into arithmetic, or letting a group skip a decision.

Case key (facilitator only)

The case was designed so that more than one explanation stays plausible. Of the 23 cases, **9** worked at one of the plants (6 Keller, 3 Ashby), so occupational exposure is plausible. **4** more lived in the Flats while Keller operated but never worked there, so historical residential exposure is possible but unproven. **10** have no identifiable link, consistent with background cases (age, smoking, chance). Today's concentration in the Flats is partly produced by Riverside Commons (5 older residents who moved in 2016–2020).

Case	Home at diagnosis	Age, year	Home in 1985	Worked at	Possible link	History
F1	Flats (Commons)	78, 2022	Westend	Keller 1968-1986	Work: Keller	Lived in Westend 1966-2017; moved to Riverside Commons 2017
F2	Flats (Commons)	81, 2021	Out of town		None identified	Lived in New Jersey until 2016; moved to Riverside Commons to be near family
F3	Flats (Commons)	74, 2023	Hillcrest		None identified	Lived in Hillcrest 1979-2018; moved to Riverside Commons 2018
F4	Flats (Commons)	79, 2024	Out of town		None identified	Lived in a town 40 miles away until 2019; moved to Riverside Commons 2019
F5	Flats (Commons)	83, 2025	Westend		None identified	Lived in Westend 1970-2020; moved to Riverside Commons 2020
F6	Flats	80, 2021	Flats	Keller 1962-1990	Work: Keller	Same Flats house since 1961
F7	Flats	73, 2023	Flats	Keller 1969-1994	Work: Keller	Born in the Flats in 1950; never moved
F8	Flats	72, 2024	Flats	Keller 1975-1992	Work: Keller	Moved to the Flats in 1975 for a job at Keller
F9	Flats	77, 2022	Flats		Lived in Flats while plant ran	Same Flats house since 1958; worked in a downtown store
F10	Flats	66, 2025	Flats		Lived in Flats while plant ran	Grew up in the Flats from 1970; elementary school teacher
F11	Flats	70, 2022	Northside	Ashby 1985-2009	Work: Ashby	Lived in Northside 1980-1999; moved to the Flats 1999
F12	Flats	64, 2023	Northside		None identified	Grew up in Northside; moved to the Flats 2004
F13	Flats	59, 2025	Out of town		None identified	Lived out of state until 2011
D1	Downtown	75, 2021	Northside	Keller 1972-1990	Work: Keller	Lived in Northside until 1990; then Downtown
D2	Downtown	71, 2024	Flats		Lived in Flats while plant ran	Grew up and lived in the Flats 1953-1990; then Downtown
D3	Downtown	62, 2023	Out of town		None identified	Lived out of state until 2015
W1	Westend	74, 2022	Westend	Ashby 1979-2005	Work: Ashby	Same Westend house since 1979
W2	Westend	69, 2025	Out of town		None identified	Lived in another county until 1988
N1	Northside	68, 2023	Flats		Lived in Flats while plant ran	Grew up and lived in the Flats 1955-1990; then Northside
N2	Northside	76, 2024	Out of town		None identified	Lived out of state until 2009
H1	Hillcrest	69, 2023	Flats	Keller 1976-1990	Work: Keller	Frank: grew up in the Flats; lived there until 2001; then Hillcrest
H2	Hillcrest	67, 2024	Westend	Ashby 1990-2009	Work: Ashby	Lived in Westend 1980-1995; then Hillcrest
H3	Hillcrest	72, 2021	Out of town		None identified	Lived out of state until 2007